

Research Project Proposal

District-Level Dengue Outbreak Early Warning System for Sri Lanka

Group Number: Group 10

Group Members:

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1 Introduction

Dengue fever is a severe, mosquito-borne viral disease that has evolved into a hyper-endemic public health crisis in Sri Lanka. The virus is primarily transmitted by *Aedes aegypti* and *Aedes albopictus* mosquitoes, and its spread is closely tied to our country's weather, especially the Southwest and Northeast monsoons. At present, the national strategy for managing this illness is largely retrospective, depending on hospital records of already infected individuals rather than anticipating future outbreak locations.

Shifting toward a forward-looking strategy is essential for protecting communities and optimizing medical supplies. Advanced computational techniques, specifically Machine Learning (ML), offer the ability to anticipate future surges by evaluating environmental indicators such as rainfall and temperature (Uduwanage et al., 2025). Additionally, tracking how populations move between different areas can significantly enhance the precision of these digital early warning tools in mapping disease propagation (Chen & Moraga, 2026).

Driven by the serious impact of these outbreaks on local communities, this project sets out to develop an innovative software solution that overcomes the limitations of existing methods, which frequently falter during unexpectedly severe epidemic spikes. The goal is to engineer a comprehensive dual-stream digital system that simultaneously estimates future patient volumes and flags imminent high-risk periods, ultimately delivering a practical, 4-week-ahead forecasting dashboard to support public health authorities.

2 Problem Statement

2.1 The Core Challenge: Reactive vs. Proactive

Dengue management in Sri Lanka is largely reactive, triggering interventions only after infection rates spike. While countries like Singapore and Brazil use Machine Learning (ML) to anticipate outbreaks, Sri Lanka lacks a localized, proactive early warning system that provides the necessary lead time for effective prevention.

2.2 Key Gaps and Challenges

- **Data Fragmentation:** Existing models rely primarily on rainfall, failing to integrate multidimensional factors like Human Mobility and PHI Larval Surveillance data for high-accuracy predictions.
- **Geographical Oversimplification:** Literature often overlooks the distinct dengue-climate dynamics between Wet-Zone and Dry-Zone districts and the separate impacts of the Southwest and Northeast monsoons.

- **Lead-Time Inefficiency:** Without a district-specific lag analysis, Public Health Inspectors (PHIs) cannot time their vector control interventions precisely, resulting in inefficient resource allocation.
- **Technical Limitations:** Traditional models (e.g., ARIMA) struggle with the non-linear temporal patterns of long-term data. Advanced architectures like LSTM and XGBoost are essential to model these complex 7-year longitudinal relationships.

2.3 Proposed Solution: Actionable Prevention

This research introduces a Dual-Stream Intelligence framework integrating human mobility and PHI larval reports with micro-climatic data. By utilizing LSTM and XGBoost architectures, the system converts complex epidemiological patterns into a 4-week-ahead predictive dashboard. This tool transforms raw data into actionable insights, empowering health authorities to transition from crisis management to proactive strategic prevention.

3 Objectives

3.1 Primary Objective

The primary goal of this research is to develop a proactive, data-driven dengue early warning and risk-mapping system for Sri Lanka. By leveraging automated surveillance, the project aims to empower health authorities to identify potential outbreaks in advance. This ensures timely interventions and optimized resource allocation to significantly mitigate public health risks across the country.

3.2 Secondary Objectives

1. **Data Fusion:** Integrate historical cases, climate, mobility, and PHI larval data into a single unified framework.
2. **Mobility Modeling:** Incorporate bus/train network patterns to predict inter-district disease spread.
3. **Dual-Function ML:** Develop hybrid models for both case forecasting (Regression) and risk-level categorization (Classification).
4. **Regional Nuance:** Capture Wet/Dry zone variations and establish district-specific climate-to-outbreak time lags.
5. **Decision Support:** Build a digital dashboard to translate technical forecasts into prioritized field tasks for PHIs.

Expected Impact:

- **Early Intervention:** Enables a “4-week lead time” to stop outbreaks before they escalate.
- **Optimized Resources:** Directs cleanup and inspection teams to high-risk clusters with precision.
- **Policy Support:** Provides an evidence-based, scalable framework for national data-driven healthcare.

4 Preliminary Literature Review

4.1 Existing Research and Key Findings

Dengue forecasting in Sri Lanka has been a significant area of study, with research consistently linking disease outbreaks to climatic factors such as rainfall, temperature, and humidity (Wickramathilaka & Munasinghe, 2024). These environmental variables directly influence the breeding cycles and population density of the *Aedes* mosquito (Withanage et al., 2018).

- **Methodologies:** Recent research has shifted toward Deep Learning techniques, particularly Long Short-Term Memory (LSTM) networks. These models are favored for their ability to handle non-linear patterns and long-term dependencies in historical data (Bannoud et al., 2026; Uduwanage et al., 2025).
- **Strengths:** Current forecasting models effectively demonstrate the correlation between weather fluctuations and immediate spikes in dengue cases, providing a foundation for health sectors (Karasinghe et al., 2024).
- **Weaknesses:** However, much of the existing research remains theoretical and focused on model accuracy rather than practical use. There is a notable lack of functional software dashboards that health authorities can use in real-time (Wickramathilaka & Munasinghe, 2024). Additionally, many studies focus primarily on the Western Province, which may not accurately reflect the disease dynamics in other ecological zones of the country (Fernando et al., 2024).

4.2 Research Gap and Justification

A review of prior literature highlights a critical gap in turning high-accuracy predictive models into actionable, software-driven tools for the public health sector (Wickramathilaka & Munasinghe, 2024).

- **Addressing the Gaps:** This research addresses these limitations by developing a Functional Predictive Dashboard. To move beyond the common "Western Province Bias," our study analyzes one critical district per province, ensuring the tool is relevant and accurate for the entire country (Fernando et al., 2024).
- **Justification:** This study is justified by the urgent need for proactive forecasts that assist in efficient resource allocation during outbreaks (Rahulan et al., 2024). By utilizing a weekly panel dataset since 2017, we diverge from climate-only models by also incorporating human mobility and PHI larval reports, which are crucial but often overlooked factors in dengue transmission (Del-Águila-Mejía et al., 2024).

5 Methodology

5.1 Research Approach

This study utilizes a Quantitative Modeling and System Development approach. We focus on integrating multi-source data through a hybrid machine learning design to enable proactive dengue surveillance across all nine provinces. This strategy addresses the "Western Province Bias" by incorporating localized ecological dynamics from across the country (Fernando et al., 2024).

5.2 Data Strategy and Implementation Plan

The research follows a five-phase execution plan to transition from multi-source data collection to a functional software output:

1. Data Collection:

- **Epidemiological:** Weekly cases from the Epidemiology Unit (2017–2026) (Wickramathilaka & Munasinghe, 2024).
- **Environmental:** Rainfall, temperature, and humidity from the Department of Meteorology (Withanage et al., 2018).
- **Mobility & Larval:** Commuter flow from transmit data and PHI reports (Karasinghe et al., 2024).

2. Preprocessing & Feature Engineering:

Data will be merged using Python (Pandas). Engineering focus includes –8 week weather lags, Zone Indicators (Wet/Dry zones), and Transit Risk Indices to capture regional transmission dynamics.

3. Modeling:

Comparison of LSTM (temporal patterns), LightGBM, and XGBoost (spatial variations) to identify the most robust forecasting architecture (Bannoud et al., 2026; Uduwanage et al., 2025).

4. **Evaluation:** Models will be validated using a Walk-forward validation approach for 4-week lead-time performance.
 - **Metrics:** Accuracy is measured via regression (MAE, RMSE, MAPE) and time-series (forecast bias) metrics.
 - **Outbreak Capability:** Classification metrics (Precision, Recall, F1-score, ROC-AUC) will be used against defined thresholds, with high priority on Recall to minimize missed outbreaks.
 - **Spatial Accuracy:** Comparing predicted versus actual hotspots across the nine provinces.
5. **Implementation:** Developing a React.js dashboard featuring spatial risk maps with Hotspot Marking and predictive trend alerts for outbreak prevention.

6 Project Schedule

6.1 Gantt Chart

The project timeline from initiation to completion is visualized below:

Milestone Description	April	May	June	July	August	September	October
Proposal Submission & Presentation	Kavindu Hiruni Mihiran Sachini						
Data Collection & Preprocessing		Mihiran Kavindu					
Feature Engineering & Selection		Hiruni Sachini					
Interim Reporting			Kavindu Hiruni				
Model Development & Training			Kavindu Hiruni Mihiran Sachini	Kavindu Hiruni Mihiran Sachini			
Evaluation & Validation				Hiruni Sachini			
Dashboard Development				Kavindu Mihiran	Kavindu Mihiran		
Testing & Paper Writing					Mihiran Sachini	Mihiran Sachini	
Final Thesis Writing			Kavindu Hiruni Sachini Mihiran	Kavindu Hiruni Sachini Mihiran	Kavindu Hiruni Sachini Mihiran	Kavindu Hiruni Sachini Mihiran	Kavindu Hiruni Sachini Mihiran

Table 1: Project Milestone Schedule with Team Responsibility

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